This WACV workshop paper is the Open Access version, provided by the Computer Vision Foundation.
Except for this watermark, it is identical to the accepted version;
the final published version of the proceedings is available on IEEE Xplore.

RAVEN: A Rapid Agentic Vision Framework for Emergency Response in Vulnerable Settlements

Arkadip Maitra
Red Hat
Bengaluru, India
amaitra@redhat.com

Suvajit Patra
RKMVERI
Belur, India
suvajit.patra.cs20@gm.rkmvu.ac.in

Sumantro Mukherjee
Red Hat
Bengaluru, India
sumantro@redhat.com

Abstract

*Escalating global conflicts and climate-induced displacement have led to unprecedented growth in refugee populations, with settlements often forming rapidly in response to emergencies. Traditional manual assessment methods cannot keep pace with this scale: a single refugee camp can span over 20 km², containing tens of thousands of structures that require evaluation for resource allocation, infrastructure planning, and aid distribution. Without automated analysis capabilities, humanitarian organizations face critical delays in understanding settlement conditions and responding to urgent needs. We introduce **RAVEN** (**R**apid **A**gentic **V**ision fram**E**work for emerge**N**cy response), an agentic framework that automates refugee settlement assessment from aerial and satellite imagery. RAVEN employs a Large Language Model (LLM) based orchestrator to coordinate specialized computer vision tools—including segmentation models for roof and solar panel detection, and classification models for roof material and sanitary facility identification—to generate comprehensive rapid assessment reports. These reports provide actionable insights on weather vulnerability, health risks, and energy access. We evaluate RAVEN on the KAKUMA-AERIAL dataset. The segmentation model achieves 0.922 IoU for buildings and 0.887 IoU for solar panels; the classification models achieve 86.17% accuracy for roof material and 98.20% accuracy for sanitary facility detection. In expert evaluation, the generated rapid response reports receive an average quality score of 8.63/10, demonstrating RAVEN's potential to significantly accelerate humanitarian response operations.*

1. Introduction

Escalating global conflicts, climate-induced disasters, and mass displacement events have driven unprecedented growth in refugee populations worldwide. The United Nations High Commissioner for Refugees (UNHCR) estimates that over 100 million people are currently forcibly displaced, with settlements forming rapidly—often in remote or conflict-affected regions where field access is severely limited [2, 7]. Effective humanitarian response requires timely, accurate assessment of settlement conditions: How many shelters exist? What is the quality of roofing materials? Are sanitation facilities adequate? Is there access to electricity? Answering these questions at scale remains a critical bottleneck for aid organizations.

A single refugee camp can span tens of kilometers, that require evaluation. Traditional manual assessment methods—relying on field surveys and human image interpretation—cannot keep pace with this scale. Early computer vision approaches demonstrated the feasibility of extracting dwelling footprints and camp extents from satellite imagery [26], but focused primarily on structural detection rather than the deeper "livability" features essential for humanitarian planning.

The advent of deep learning has significantly advanced remote sensing capabilities. Convolutional neural networks (CNNs) have been shown to extract meaningful socioeconomic and structural information from high-resolution imagery [12], while disaster-mapping works demonstrated reliable detection of building damage and crisis indicators [3, 27]. Advances in hyperspectral analysis [11, 18] and learned representations [10, 20] have further enabled capture of subtle material and environmental cues. However, large-scale datasets such as xView [13] and fMoW [5] primarily target developed infrastructure, while efforts like Open Buildings [23] provide footprints but lack the granularity to characterize shelter quality, sanitation, or energy access in informal settlements.

Recent work has begun addressing this gap. The KakumaAerial dataset [8] introduced comprehensive benchmarks for humanitarian machine learning, including building and solar-panel segmentation, roof-material classification, and toilet identification. Transformer-based architectures such as SegFormer [29] and contrastive models like

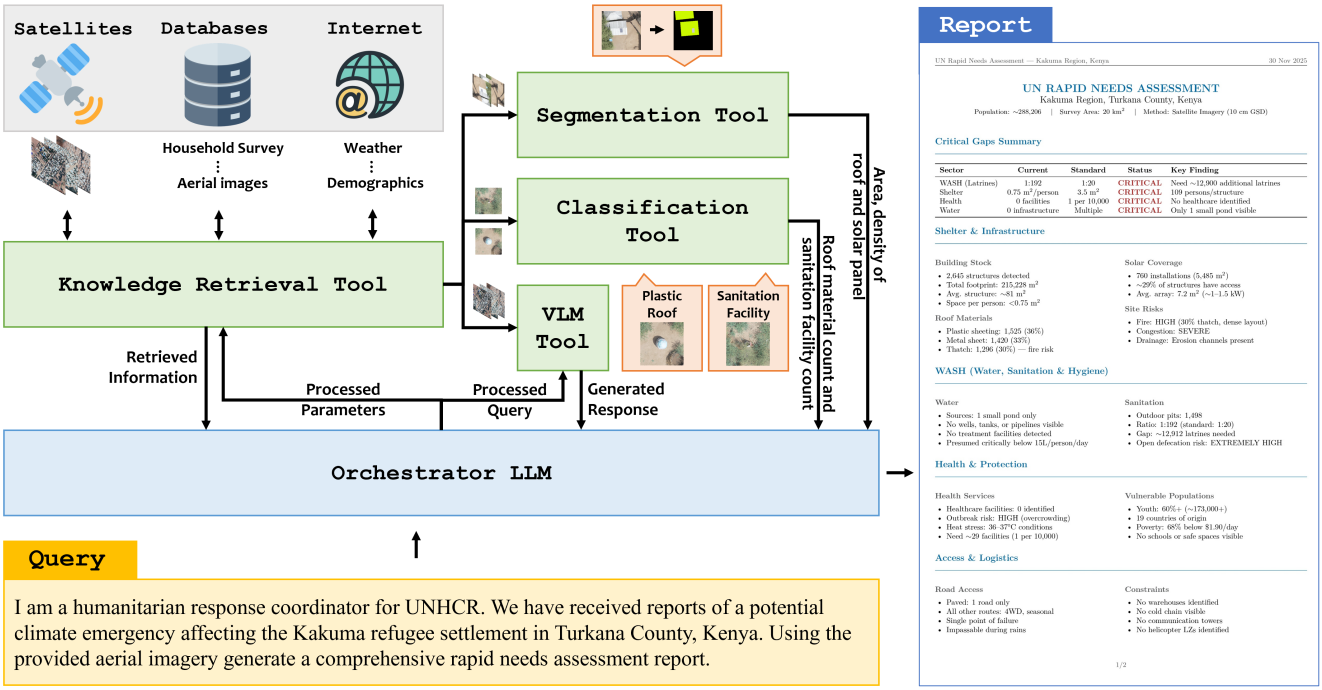


Figure 1. **RAVEN** An LLM-based orchestrator coordinates four specialized tools—Segmentation, Classification, VLM, and Knowledge Retrieval—to generate rapid assessment reports from aerial imagery and contextual data.

RemoteCLIP [15] have achieved strong performance on these tasks. Yet, translating these technical capabilities into actionable humanitarian intelligence requires more than isolated model outputs—it demands an integrated system that can orchestrate multiple analysis tools and synthesize their results into coherent, decision-ready reports.

Concurrently, the evolution of Large Language Models (LLMs) has enabled autonomous agentic frameworks capable of reasoning and tool orchestration [30]. Systems like HuggingGPT [22] and Visual ChatGPT [28] demonstrated LLMs acting as controllers that delegate tasks to specialized models, while Vision-Language Models (VLM)s such as LLaVA [16] and Qwen-VL [1] have integrated visual understanding directly into language models. However, the application of hierarchical agentic workflows to large-scale geospatial humanitarian analysis remains largely unexplored.

In this paper, we introduce **RAVEN**, an end-to-end agentic framework for automated refugee settlement assessment. To the best of our knowledge, RAVEN is the first agentic framework that integrates geospatial computer vision tools with LLM-based orchestration specifically for humanitarian settlement analysis.

Our contributions are as follows:

1. **Novel agentic framework for humanitarian assessment.** We introduce RAVEN, the first agentic system that orchestrates specialized segmentation, classification, vision-language, and knowledge retrieval tools under a unified LLM controller to automate refugee settlement assessment—bridging the gap between isolated model outputs and actionable humanitarian intelligence.
2. **State-of-the-art performance on humanitarian benchmarks.** We advance performance on the KakumaAerial dataset, achieving 0.922 IoU for building segmentation, 0.887 IoU for solar panel detection, 86.17% accuracy for roof-material classification, and 98.20% accuracy for sanitary facility identification.
3. **Scalable, model-agnostic architecture.** RAVEN's modular design supports diverse LLM backends—from local models (Qwen2.5-3B [19]) for offline deployment to cloud APIs (Claude-4.5-Opus[1]) for maximum capability—enabling deployment across resource-constrained field operations and well-equipped command centers alike.
4. **End-to-end report generation with expert validation.** We demonstrate that RAVEN generates comprehensive rapid assessment reports that receive an average expert quality score of 8.63/10, validating its practical utility for humanitarian decision-making.

[1]https://platform.claude.com/docs/en/about-claude/models/overview

2. Method

This section presents our refugee settlement analysis framework in detail. We first define the task formally, then describe the RAVEN architecture and its constituent components: the main orchestrator LLM, segmentation, and classification models, knowledge retrieval tool, and the VLM tool.

2.1. Task Definition

Given a target refugee settlement $\mathcal{S}$ defined by a bounding polygon $\mathcal{B} = \{(lat_i, lon_i)\}_{i=1}^{m}$ and a collection of aerial images $\mathcal{I} = \{I_1, \ldots, I_n\}$ covering the settlement area, our framework produces a structured rapid assessment report [24] to support humanitarian response coordination.

2.2. RAVEN: A Rapid Agentic Vision Framework for Emergency Response in Vulnerable Settlements

We present **RAVEN** (**R**apid **A**gentic **V**ision fram**E**work for emerge**N**cy response), a multi-agent framework for automated analysis of refugee camps and informal settlements. RAVEN leverages a central LLM-based orchestrator that coordinates specialized tools—including vision-language models for aerial image interpretation, segmentation models for roof and infrastructure detection, and classification models for settlement characterization—to generate rapid response reports for emergency humanitarian efforts.

Figure 1 illustrates the high-level architecture. At the core lies a central reasoning agent that interprets user queries ranging from natural language questions and dynamically invokes the appropriate combination of tools to construct comprehensive responses. The system supports information retrieval from trusted online sources and leverages aerial images and satellite imagery to produce detailed situational assessments.

RAVEN's LLM-based orchestrator coordinates four primary tools. The *Segmentation Tool* analyzes aerial imagery to detect roofs and solar panels, computing their coverage area using image metadata (e.g., cm/pixel resolution). The *Classification Tool* identifies roof types and sanitary facilities, returning counts for each category. The *Knowledge Retrieval Tool* queries local documentation and trusted online sources to gather contextual information. Finally, the *VLM Tool* (Vision-Language Model) is dynamically invoked when additional visual information is required for answering ad-hoc queries.

The key design principles guiding RAVEN's architecture are:

1. **Query-driven orchestration**: The orchestrator determines which analytical tools are necessary based on the user's information needs.
2. **Multi-source data fusion**: Information from aerial imagery, external APIs, and web sources is synthesized into coherent responses.
3. **Graceful degradation**: When primary data sources are unavailable or incomplete, the system employs fallback strategies to maximize response utility.
4. **Modular adaptability**: Every component in the pipeline—from the core LLM to individual computer vision models—can be independently swapped or upgraded without affecting the overall system architecture.

2.3. Orchestrator LLM

The central orchestrator in RAVEN is powered by a LLM responsible for query interpretation, tool selection, and response synthesis. A key design decision is *model agnosticism*—the framework does not depend on any single LLM provider, allowing operators to select the most appropriate model based on available resources and deployment constraints.

RAVEN supports a range of LLMs spanning different capability tiers and deployment modes:

- **Local models**: Qwen2.5-3B [19] and Qwen2.5-32B [19] for offline or resource-constrained deployments where network connectivity is limited, or data privacy is a concern.
- **Cloud-hosted models**: Gemini-3-Pro[2], Claude-4.5-Opus, and GPT-5.1[3] for scenarios requiring maximum reasoning capability and where reliable internet access is available.

This flexibility is particularly important in humanitarian contexts, where deployment conditions vary widely. Field operations may face intermittent connectivity, limited computational infrastructure, or data sovereignty requirements that preclude cloud-based inference.

While any compatible LLM can serve as the orchestrator, our experimental results (Section 3) demonstrate that Claude-4.5-Opus achieves the best performance on orchestration tasks, particularly in tool selection accuracy and response coherence 4.

2.4. Segmentation Tool

We employ SegFormer [29], a hierarchical Transformer encoder with a lightweight All-MLP decoder for semantic segmentation. The encoder produces multi-scale feature maps $\{F_1, F_2, F_3, F_4\}$ at resolutions $\{\frac{H}{4}, \frac{H}{8}, \frac{H}{16}, \frac{H}{32}\}$. The decoder unifies these features through linear projections and upsampling:

$$F = \text{MLP}\left(\text{Concat}\left(\text{Upsample}(\hat{F}_i)\right)_{i=1}^{4}\right) \quad (1)$$

2.4.1. Loss Function

To address class imbalance, we combine Focal Loss and Dice Loss:

[2] https://ai.google.dev/gemini-api/docs
[3] https://platform.openai.com/docs/models/gpt-5.1/

$$\mathcal{L}_{\text{total}} = \lambda_f \cdot \mathcal{L}_{\text{focal}} + \lambda_d \cdot \mathcal{L}_{\text{dice}} \tag{2}$$

Focal Loss [14] down-weights well-classified examples:

$$\mathcal{L}_{\text{focal}} = -\sum_{c=1}^{C} \alpha_c (1 - p_c)^{\gamma} \log(p_c) \tag{3}$$

where α_c are inverse-frequency class weights and γ is the focusing parameter.

Dice Loss [17] optimizes region overlap, improving segmentation of small objects:

$$\mathcal{L}_{\text{dice}} = 1 - \frac{1}{C} \sum_{c=1}^{C} \frac{2 \sum_i p_{c,i} \cdot g_{c,i} + \epsilon}{\sum_i p_{c,i} + \sum_i g_{c,i} + \epsilon} \tag{4}$$

2.4.2. Output Metrics

The Segmentation Tool converts pixel-level predictions into physical measurements using image metadata. Given the ground sampling distance (GSD) in cm/pixel, the tool computes:

- **Roof coverage area** (A_{roof}): Total area covered by detected roof structures in square meters.
- **Solar panel area** (A_{solar}): Total area covered by solar installations in square meters.
- **Settlement density**: Ratio of built-up area to total settlement area, providing an indicator of shelter crowding.
- **Solar adoption rate**: Ratio of solar panel area to roof area ($A_{\text{solar}}/A_{\text{roof}}$) providing an indicator of electricity production.

The area conversion is computed as:

$$A_c = N_c \times \left(\frac{\text{GSD}}{100} \right)^2 \quad (\text{m}^2) \tag{5}$$

where N_c is the pixel count for class c and GSD is the ground sampling distance in cm/pixel.

2.4.3. Evaluation Metrics

Pixel-level: We report Intersection over Union (IoU), precision, recall, and F1 score per class:

$$\text{IoU}_c = \frac{|P_c \cap G_c|}{|P_c \cup G_c|}, \quad \text{mIoU} = \frac{1}{2} \sum_{c \in \{\text{bld,sol}\}} \text{IoU}_c \tag{6}$$

Predictions are optionally refined through morphological closing, hole filling, and small object removal to improve object-level metrics.

2.5. Classification Tool

We employ RemoteCLIP [15], a contrastive language-image model pretrained on large-scale remote sensing imagery. Unlike conventional classification approaches that employ a task-specific classification head, our method leverages CLIP-style zero-shot transfer by computing similarities between image embeddings and text embeddings representing class descriptions.

2.5.1. Output Metrics

The Classification Tool processes individual structure patches extracted from the segmentation output and returns categorical counts that inform humanitarian assessment:

- **Roof type distribution**: Counts of structures by roofing material (e.g., metal sheet, tarpaulin, thatch, concrete). This serves as a proxy for *weather resistivity*—settlements with high proportions of tarpaulin or thatch roofs are more vulnerable to heavy rainfall, wind damage, and extreme temperatures.
- **Sanitary facility count**: Number of detected latrines and sanitation structures. Low sanitary facility density relative to shelter count indicates elevated *health risks*, including potential for waterborne disease outbreaks and inadequate waste management.

2.6. Knowledge Retrieval Tool

The Knowledge Retrieval Tool queries online APIs for geolocation data, weather conditions, and demographic information from humanitarian databases, and can perform web searches for recent news and situation reports. For offline deployments or sensitive data, the tool ingests local files in common formats. When invoked, it receives the necessary parameters from the orchestrator, selects the appropriate source, and returns a structured response with source attribution for transparency.

2.7. VLM Tool

The VLM Tool provides general-purpose visual reasoning for queries beyond the scope of specialized tools. It is dynamically invoked when users request damage assessment, landmark identification, scene descriptions, or other ad-hoc visual analysis. For instance, queries asking about "flooding" or "road condition" are routed to the VLM Tool.

In our implementation, we use Qwen3-VL-8B-Instruct [1] as the default VLM. Upon invocation, the VLM Tool receives multiple images and a natural language prompt, returning a textual response that the orchestrator incorporates into the final assessment.

3. Experiments

In this section, we first evaluate the core perception components of our framework, including the building and solar-panel segmentation model, the roof-material classification model, and the toilet-classification model. We then assess several LLMs as orchestrator agent , analyzing both their ability to coordinate the available tools and their overall effectiveness in executing the end-to-end assessment workflow. All the generated rapid response reports can be found in the supplementary material.

Table 1. Perofrmance of Roof and Solar Segmentation. (Top is previous reported baseline results and bottom is our runs).

Method	Variant / Details	Pixel IoU		Object Precision		Object Recall	
		Build.	Solar	Build.	Solar	Build.	Solar
Baseline Results [8]							
U-Net	ResNet-50	0.777	0.735	0.622	0.422	0.622	0.700
U-Net	ResNeXt-50	0.812	0.772	0.677	0.500	0.832	0.700
DeepLabV3+	ResNet-50	0.772	0.744	0.664	0.318	0.692	0.671
DeepLabV3+	ResNeXt-50	0.815	0.756	0.703	0.325	0.772	0.643
DeepLabV3+	EfficientNet B3	0.775	0.167	0.705	0.013	0.742	0.607
U-Net EfficientNet B3	Jaccard	0.848	0.813	0.830	0.615	0.822	**0.764**
U-Net EfficientNet B3	Cross Entropy	0.851	0.711	0.802	0.255	0.878	0.536
U-Net EfficientNet B3	Focal	0.830	0.790	0.743	0.777	0.549	0.600
U-Net EfficientNet B3	Buffered + Jaccard	0.851	0.774	0.874	0.396	0.846	0.721
U-Net EfficientNet B3	Eroded-Buffered + Jaccard	0.830	0.787	0.870	0.500	0.824	0.700
Ours (SegFormer Runs)							
SegFormer-B1	Focal	0.908	0.853	0.820	0.758	0.878	0.632
SegFormer-B1	Focal + Dice	0.913	0.851	0.880	0.752	**0.886**	0.612
SegFormer-B1	Focal + Dice + Multi-scale	0.919	0.884	**0.887**	0.756	0.863	0.600
SegFormer-B2	Weighted CE + Multi-scale	**0.922**	0.882	0.871	0.764	0.866	0.671
SegFormer-B2	Focal + Dice	0.918	**0.887**	0.873	**0.778**	0.862	0.728

Table 2. Performance for Roof Material Classification.

Model	Configuration	Eval	Accuracy	Precision	Recall	F1-Score
Baseline [8]	ResNet-18 Variant	All	85.59	–	–	–
RemoteCLIP ResNet-50	Strong aug + focal	All	84.77	86.60	84.77	84.98
		w/o Other	85.17	87.06	85.17	85.38
RemoteCLIP ViT-B-32	Strong aug + focal	All	84.30	86.45	84.30	84.74
		w/o Other	84.58	86.78	84.58	85.02
RemoteCLIP ViT-L-14	Weak aug + focal	All	85.77	88.08	85.77	86.21
		w/o Other	**86.17**	**88.54**	**86.17**	**86.62**

Table 3. Performance for Sanitary Facility Detection.

Model	Best Configuration	Accuracy
Baseline [8]	ResNet-18 Variant	97.8%
ResNet-50	Focal Loss + Weights	97.00%
ViT-B-32	Focal Loss + Weights	96.60%
ViT-L-14	Focal Loss + Weights	**98.20%**

3.1. Dataset

We utilize the **KakumaAerial** dataset [8], a benchmark dataset featuring imagery from the Kakuma and Kalobeyei refugee camps in Turkana County, Kenya. The KakumaAerial dataset was selected for its exceptionally high-resolution aerial imagery and its well-structured pre-chipped formats that facilitate efficient training of both segmentation and classification models.

The dataset provides pre-chipped subsets at 256×256 pixel resolution, specifically designed for training deep learning models.[4] The dataset comes partitioned at the tile level to prevent spatial leakage between training and evaluation sets.

3.2. Building and Solar Panel Segmentation

Table 1 summarizes the quantitative performance across different architectures, backbones, and loss functions. Among the baseline models, U-Net [21] with EfficientNet-B3 [25] consistently outperformed both ResNet-50 [9] and ResNet-50 backbones, achieving pixel IoU scores of 0.848 and 0.813 for buildings and solar panels, respectively.

[4] https://zenodo.org/records/14727217

Table 4. Qualitative comparison of different generated rapid assessment reports

	Information Accuracy		Correct Inference		Text Quality	
Model	Grok	Human	Grok	Human	Grok	Human
Qwen-2.5-3B-Instruct	5.6	5.3	6.3	6.7	5.9	6.4
Qwen-2.5-32B-Instruct	6.4	6.6	6.9	7.3	7.0	6.9
Gemini-3-pro	8.7	8.2	8.1	7.8	8.4	**8.5**
GPT-5.1	8.9	8.2	8.8	**8.3**	8.2	8.0
Claude-4.5-Opus	**8.9**	**8.3**	**9.1**	8.1	**8.9**	**8.5**

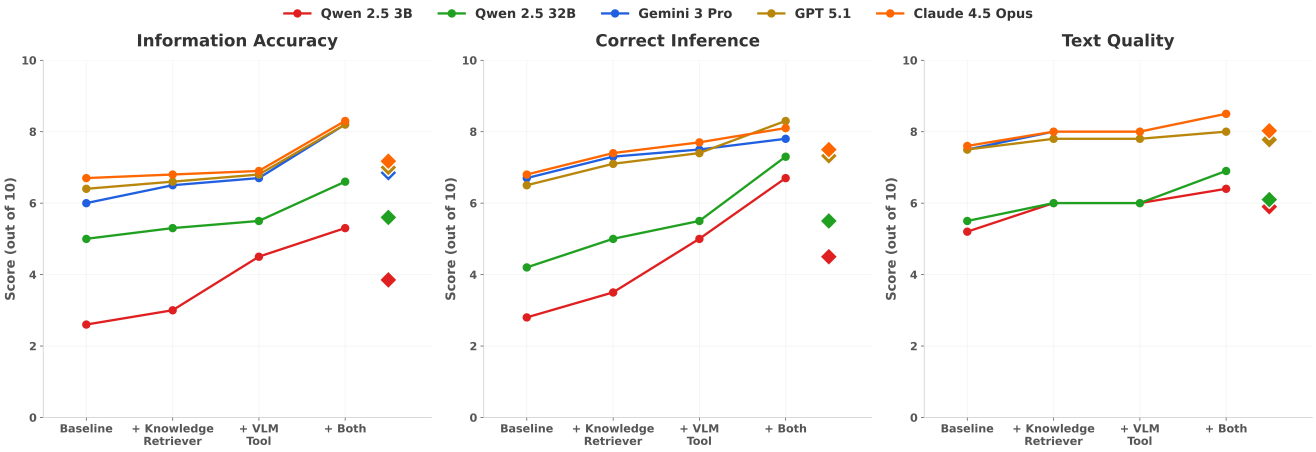


Figure 2. Effectiveness of different tools in RAVEN

DeepLabV3+ [4] showed competitive building segmentation but struggled with solar panel detection, particularly with the EfficientNet-B3 backbone (Solar IoU: 0.167).

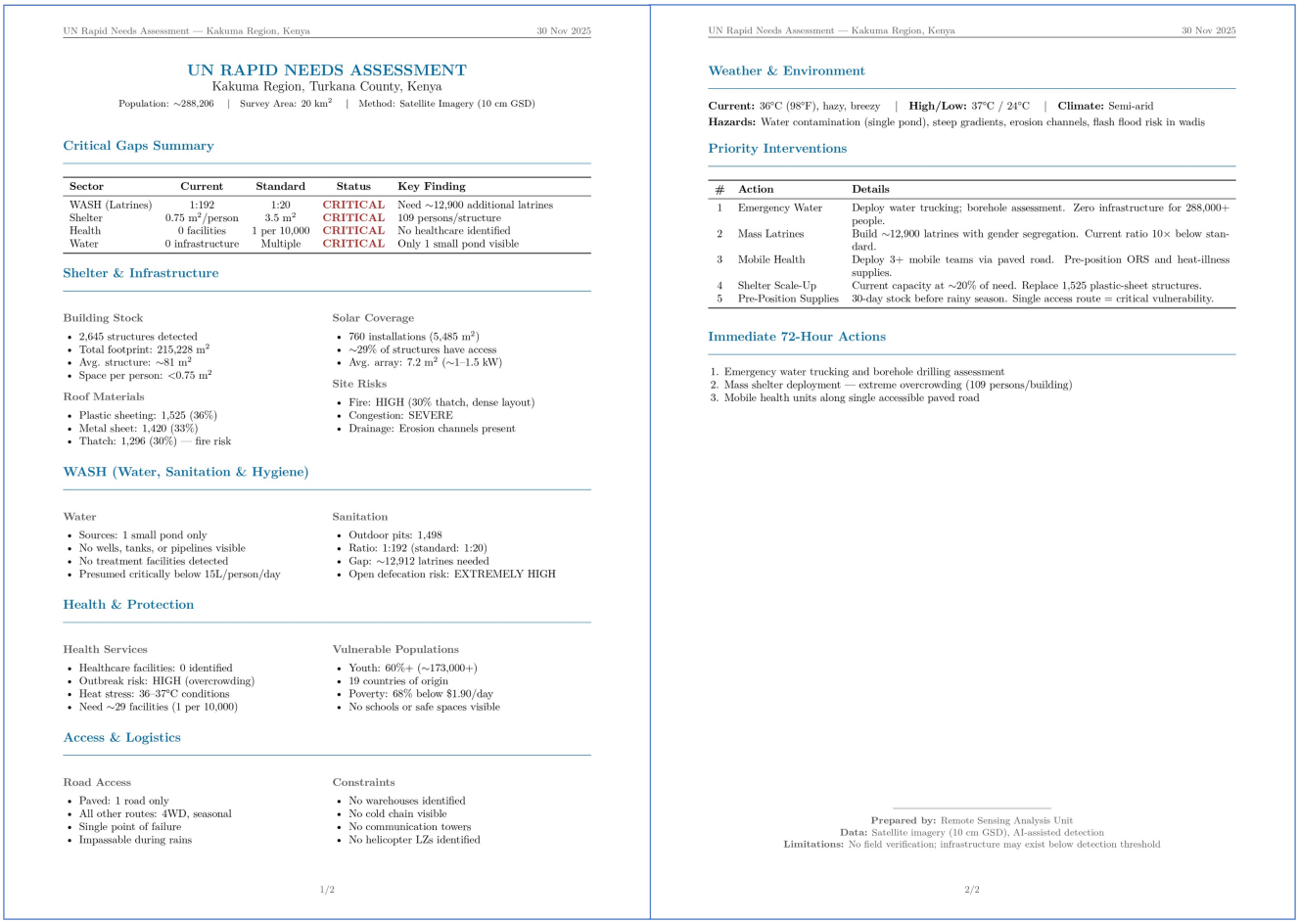
We apply the SegFormer [29] model for this task, using its ImageNet-pretrained weights, and fine-tune it for 50 epochs to adapt it to the downstream segmentation task. For our SegFormer experiments, we employed the default (unmodified) ground truth masks, as this representation most closely reflects real-world annotation practices and ensures consistency with standard evaluation protocols. SegFormer-B2 with Focal + Dice loss achieved the best overall performance, attaining the highest solar panel IoU (0.887) and object-level precision (0.778), while maintaining competitive building segmentation (IoU: 0.918). SegFormer-B2 with weighted cross-entropy and multi-scale supervision achieved the highest building IoU (0.922). Notably, SegFormer models improved building IoU by approximately 7 percentage points and solar IoU by 7.4 percentage points over the best U-Net baseline, demonstrating the effectiveness of the transformer-based architecture for this multi-class segmentation task.

3.3. Roof Material Classification

For the roof-material classification task, we use the RemoteCLIP [15] model, which was pretrained on a large-scale remote sensing image–text dataset, and fine-tune it on our task for 20 epochs. We evaluate three backbone architectures (ResNet-50 [9], ViT-B-32 [6], and ViT-L-14 [6]) with different training configurations, comparing against the baseline accuracy of 85.6% reported in the original paper [8], as listed in Table 2.

Our best model, ViT-L-14 with weak augmentation and focal loss, achieves 85.77% accuracy on the full test set and 86.17% when excluding the "other" class, with corresponding F1-scores of 86.21% and 86.62%. Focal loss consistently improves performance over cross-entropy across all architectures, with accuracy gains of 1.6–2.4 percentage points.

The "other" class remains challenging for all models due to its heterogeneous nature. When excluding this class, our fine-tuned ViT-L-14 [6] surpasses the baseline by 0.57 percentage points, demonstrating that domain-specific fine-tuning of RemoteCLIP [15] with focal loss can effectively adapt the model for roof material classification in humanitarian mapping applications.



UN Rapid Needs Assessment — Kakuma Region, Kenya 30 Nov 2025

UN RAPID NEEDS ASSESSMENT

Kakuma Region, Turkana County, Kenya

Population: ∼288,206 | Survey Area: 20 km² | Method: Satellite Imagery (10 cm GSD)

Critical Gaps Summary

Sector	Current	Standard	Status	Key Finding
WASH (Latrines)	1:192	1:20	CRITICAL	Need ∼12,900 additional latrines
Shelter	0.75 m²/person	3.5 m²	CRITICAL	109 persons/structure
Health	0 facilities	1 per 10,000	CRITICAL	No healthcare identified
Water	0 infrastructure	Multiple	CRITICAL	Only 1 small pond visible

Shelter & Infrastructure

Building Stock
- 2,645 structures detected
- Total footprint: 215,228 m²
- Avg. structure: ∼81 m²
- Space per person: <0.75 m²

Roof Materials
- Plastic sheeting: 1,525 (36%)
- Metal sheet: 1,420 (33%)
- Thatch: 1,296 (30%) — fire risk

Solar Coverage
- 760 installations (5,485 m²)
- ∼29% of structures have access
- Avg. array: 7.2 m² (∼1–1.5 kW)

Site Risks
- Fire: HIGH (30% thatch, dense layout)
- Congestion: SEVERE
- Drainage: Erosion channels present

WASH (Water, Sanitation & Hygiene)

Water
- Sources: 1 small pond only
- No wells, tanks, or pipelines visible
- No treatment facilities detected
- Presumed critically below 15L/person/day

Sanitation
- Outdoor pits: 1,498
- Ratio: 1:192 (standard: 1:20)
- Gap: ∼12,912 latrines needed
- Open defecation risk: EXTREMELY HIGH

Health & Protection

Health Services
- Healthcare facilities: 0 identified
- Outbreak risk: HIGH (overcrowding)
- Heat stress: 36–37°C conditions
- Need ∼29 facilities (1 per 10,000)

Vulnerable Populations
- Youth: 60%+ (∼173,000+)
- 19 countries of origin
- Poverty: 68% below $1.90/day
- No schools or safe spaces visible

Access & Logistics

Road Access
- Paved: 1 road only
- All other routes: 4WD, seasonal
- Single point of failure
- Impassable during rains

Constraints
- No warehouses identified
- No cold chain visible
- No communication towers
- No helicopter LZs identified

1/2

UN Rapid Needs Assessment — Kakuma Region, Kenya 30 Nov 2025

Weather & Environment

Current: 36°C (98°F), hazy, breezy | **High/Low:** 37°C / 24°C | **Climate:** Semi-arid
Hazards: Water contamination (single pond), steep gradients, erosion channels, flash flood risk in wadis

Priority Interventions

#	Action	Details
1	Emergency Water	Deploy water trucking; borehole assessment. Zero infrastructure for 288,000+ people.
2	Mass Latrines	Build ∼12,900 latrines with gender segregation. Current ratio 10× below standard.
3	Mobile Health	Deploy 3+ mobile teams via paved road. Pre-position ORS and heat-illness supplies.
4	Shelter Scale-Up	Current capacity at ∼20% of need. Replace 1,525 plastic-sheet structures.
5	Pre-Position Supplies	30-day stock before rainy season. Single access route = critical vulnerability.

Immediate 72-Hour Actions

1. Emergency water trucking and borehole drilling assessment
2. Mass shelter deployment — extreme overcrowding (109 persons/building)
3. Mobile health units along single accessible paved road

Prepared by: Remote Sensing Analysis Unit
Data: Satellite imagery (10 cm GSD), AI-assisted detection
Limitations: No field verification; infrastructure may exist below detection threshold

2/2

Figure 3. Sample rapid assessment report summary generated by Claude-4.5-Opus.

3.4. Sanitation Facilities Detection

Table 3 summarizes the classification accuracy for each backbone under its best-performing configuration. We use RemoteCLIP [15], which was pretrained on a large-scale remote sensing image–text dataset, and fine-tune it on our task for 20 epochs. The baseline accuracy of 97.8% was reported in the original paper [8].

We evaluated three RemoteCLIP backbones (ResNet-50 [9], ViT-B-32 [6], and ViT-L-14 [6]) across five training configurations combining different augmentation strategies and loss functions. Focal loss with class weights consistently yielded the best results across all backbones, indicating that addressing class imbalance is more effective than aggressive data augmentation for this task.

The ViT-L-14 backbone achieved the highest accuracy of 98.20%, surpassing the baseline by 0.4 percentage points. The smaller ResNet-50 and ViT-B-32 backbones achieved 97.00% and 96.60%, respectively, slightly below the baseline, likely due to limited model capacity during fine-tuning.

3.5. Qualitative Evaluation

To assess the quality of the outputs produced by our proposed framework, we presented the final rapid response reports (Listed in Supplementary Material) to both a human expert and the grok-4-fast-reasoning[5] model, along with a reference UN rapid response report. Each generated report was evaluated across three dimensions: *Information Accuracy*, *Correct Inference from Information*, and *Quality of Writing*. The average scores for each category, computed over 10 generated rapid response reports, are shown in Table 4.

A sample rapid response review generated by Claude-4.5-Opus is shown in Figure 3

- **Information Accuracy**: Measures how factually correct the generated content is relative to the actual information of the refugee settlement.
- **Correct Inference from Information**: Evaluates whether conclusions, interpretations, and extrapolations

[5] https://docs.x.ai/docs/models/grok-4-1-fast-reasoning

are logically valid and supported.

- **Text Quality**: Assesses the clarity, coherence, structure, and overall readability of the generated text as compared to the sample UN rapid response report.

3.6. Ablation

The orchestrator agent's performance improves significantly when augmented with the Knowledge Retrieval Tool and VLM Tool. Figure 2 shows average human-evaluated score improvements across tool configurations, when compared to the baseline (orchestrator with segmentation and classification tools).

Adding the Knowledge Retriever yields modest gains: +0.3 in information accuracy, +0.7 in correct inference, and +0.5 in text quality, suggesting factual grounding primarily enhances reasoning over fluency.

The VLM Tool produces stronger improvements: +0.7 in information accuracy, +1.2 in correct inference, and +0.5 in text quality, establishing multimodal understanding as the more impactful single enhancement.

Combining both tools achieves peak performance with synergistic effects: +2.0 in information accuracy, +2.2 in correct inference, and +1.0 in text quality, confirming that factual grounding and visual context address complementary limitations.

4. Discussions and Conclusion

We presented RAVEN, a agentic framework for automated refugee settlement assessment from aerial imagery. Our experiments demonstrate strong performance across all components: the segmentation model achieves 0.922 IoU for buildings and 0.887 IoU for solar panels, while classification models attain 86.17% accuracy for roof materials and 98.20% for sanitary facilities. Expert evaluation of generated reports (8.63/10) validates that RAVEN produces actionable humanitarian intelligence.

Our LLM comparison reveals that Claude-4.5-Opus achieves the highest orchestration quality, followed by GPT-5.1 and Gemini 3 Pro, while local models (Qwen 2.5 [19]) show lower performance. This suggests organizations with reliable connectivity should prioritize cloud-hosted models, though local deployment remains viable for offline or privacy-sensitive operations. The ablation study confirms that combining the Knowledge Retrieval and VLM tools yields the best results, validating our multi-source data fusion approach.

Several limitations remain. Object-level solar panel detection shows room for improvement (F1: 0.67), and temporal change detection is not yet supported. Future work will address these gaps and explore uncertainty quantification and interactive querying interfaces.

In conclusion, RAVEN is the first agentic framework integrating LLM orchestration with geospatial computer vision for humanitarian assessment. By enabling rapid, scalable situational awareness, RAVEN can accelerate humanitarian response and improve resource allocation for vulnerable populations. We release RAVEN as open-source to support further research at the intersection of agentic AI and humanitarian computing.

References

[1] Jinze Bai, Shuai Bai, Shusheng Yang, Shijie Wang, Sinan Tan, Peng Wang, Junyang Lin, Chang Zhou, and Jingren Zhou. Qwen-vl: A versatile vision-language model for understanding, localization, text reading, and beyond, 2023. 2, 4

[2] Philippe Bally, Jerome Bequignon, Olivier Arino, and Stephen Briggs. Remote sensing and humanitarian aid-a life-saving combination. *Esa Bulletin*, 122:36–41, 2005. 1

[3] Benjamin Bischke, Prakriti Bhardwaj, Aman Gautam, Patrick Helber, Damian Borth, and Andreas Dengel. Detection of flooding events in social multimedia and satellite imagery using deep neural networks. In *MediaEval*, 2017. 1

[4] Liang-Chieh Chen, Yukun Zhu, George Papandreou, Florian Schroff, and Hartwig Adam. Encoder-decoder with atrous separable convolution for semantic image segmentation. In *Proceedings of the European conference on computer vision (ECCV)*, pages 801–818, 2018. 6

[5] Gordon Christie, Neil Fendley, James Wilson, and Ryan Mukherjee. Functional map of the world. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 6172–6180, 2018. 1

[6] Alexey Dosovitskiy. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020. 6, 7

[7] Silvia Giada, Tom De Groeve, Daniele Ehrlich, and Pierre Soille. Information extraction from very high resolution satellite imagery over lukole refugee camp, tanzania. *International Journal of Remote Sensing*, 24(22):4251–4266, 2003. 1

[8] Amrita Gupta, Anthony Ortiz, Simone Fobi, Duncan Kebut, Seema Iyer, Rahul Dodhia, and Juan Lavista-Ferres. Mapping refugee camps with ai: A benchmark dataset and baseline models for humanitarian applications. In *Proceedings of the Winter Conference on Applications of Computer Vision*, pages 517–525, 2025. 1, 5, 6, 7

[9] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 770–778, 2016. 5, 6, 7

[10] Danfeng Hong, Naoto Yokoya, Jocelyn Chanussot, and Xiao Xiang Zhu. An augmented linear mixing model to address spectral variability for hyperspectral unmixing. *IEEE Transactions on Image Processing*, 28(4):1923–1938, 2018. 1

[11] Danfeng Hong, Lianru Gao, Jing Yao, Bing Zhang, Antonio Plaza, and Jocelyn Chanussot. Graph convolutional networks for hyperspectral image classification. *IEEE Transactions on Geoscience and Remote Sensing*, 59(7):5966–5978, 2020. 1

[12] Neal Jean, Marshall Burke, Michael Xie, W Matthew Alampay Davis, David B Lobell, and Stefano Ermon. Combining satellite imagery and machine learning to predict poverty. *Science*, 353(6301):790–794, 2016. 1

[13] Darius Lam, Richard Kuzma, Kevin McGee, Samuel Dooley, Michael Laielli, Matthew Klaric, Yaroslav Bulatov, and Brendan McCord. xview: Objects in context in overhead imagery. *arXiv preprint arXiv:1802.07856*, 2018. 1

[14] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. Focal loss for dense object detection. In *Proceedings of the IEEE International Conference on Computer Vision*, pages 2980–2988, 2017. 4

[15] Fan Liu, Delong Chen, Zhangqingyun Guan, Xiaocong Zhou, Jiale Zhu, Qiaolin Ye, Liyong Fu, and Jun Zhou. Remoteclip: A vision language foundation model for remote sensing. *IEEE Transactions on Geoscience and Remote Sensing*, 62:1–16, 2024. 2, 4, 6, 7

[16] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning, 2023. 2

[17] Fausto Milletari, Nassir Navab, and Seyed-Ahmad Ahmadi. V-net: Fully convolutional neural networks for volumetric medical image segmentation. In *2016 Fourth International Conference on 3D Vision (3DV)*, pages 565–571. IEEE, 2016. 4

[18] Saurabh Prasad and Jocelyn Chanussot. *Hyperspectral image analysis: advances in machine learning and signal processing*. Springer Nature, 2020. 1

[19] Qwen, :, An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, Huan Lin, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jingren Zhou, Junyang Lin, Kai Dang, Keming Lu, Keqin Bao, Kexin Yang, Le Yu, Mei Li, Mingfeng Xue, Pei Zhang, Qin Zhu, Rui Men, Runji Lin, Tianhao Li, Tianyi Tang, Tingyu Xia, Xingzhang Ren, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yu Wan, Yuqiong Liu, Zeyu Cui, Zhenru Zhang, and Zihan Qiu. Qwen2.5 technical report, 2025. 2, 3, 8

[20] Behnood Rasti, Danfeng Hong, Renlong Hang, Pedram Ghamisi, Xudong Kang, Jocelyn Chanussot, and Jon Atli Benediktsson. Feature extraction for hyperspectral imagery: The evolution from shallow to deep: Overview and toolbox. *IEEE Geoscience and Remote Sensing Magazine*, 8(4):60–88, 2020. 1

[21] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. In *International Conference on Medical image computing and computer-assisted intervention*, pages 234–241. Springer, 2015. 5

[22] Yongliang Shen, Kaitao Song, Xu Tan, Dongsheng Li, Weiming Lu, and Yueting Zhuang. Hugginggpt: Solving ai tasks with chatgpt and its friends in hugging face. *Advances in Neural Information Processing Systems*, 36:38154–38180, 2023. 2

[23] Wojciech Sirko, Sergii Kashubin, Marvin Ritter, Abigail Annkah, Yasser Salah Eddine Bouchareb, Yann Dauphin, Daniel Keysers, Maxim Neumann, Moustapha Cisse, and John Quinn. Continental-scale building detection from high resolution satellite imagery. *arXiv preprint arXiv:2107.12283*, 2021. 1

[24] Sphere Association. *The Sphere Handbook: Humanitarian Charter and Minimum Standards in Humanitarian Response*. Sphere Association, Geneva, Switzerland, 4th edition, 2018. 3

[25] Mingxing Tan and Quoc Le. Efficientnet: Rethinking model scaling for convolutional neural networks. In *International conference on machine learning*, pages 6105–6114. PMLR, 2019. 5

[26] Dirk Tiede, Petra Füreder, Stefan Lang, Daniel Hölbling, and Peter Zeil. Automated analysis of satellite imagery to provide information products for humanitarian relief operations in refugee camps–from scientific development towards operational services. *PFG Photogrammetrie–Fernerkundung–Geoinformation*, 3(2013):185–195, 2013. 1

[27] Anand Vetrivel, Markus Gerke, Norman Kerle, Francesco Nex, and George Vosselman. Disaster damage detection through synergistic use of deep learning and 3d point cloud features derived from very high resolution oblique aerial images, and multiple-kernel-learning. *ISPRS journal of photogrammetry and remote sensing*, 140:45–59, 2018. 1

[28] Chenfei Wu, Shengming Yin, Weizhen Qi, Xiaodong Wang, Zecheng Tang, and Nan Duan. Visual chatgpt: Talking, drawing and editing with visual foundation models. *arXiv preprint arXiv:2303.04671*, 2023. 2

[29] Enze Xie, Wenhai Wang, Zhiding Yu, Anima Anandkumar, Jose M Alvarez, and Ping Luo. Segformer: Simple and efficient design for semantic segmentation with transformers. *Advances in neural information processing systems*, 34: 12077–12090, 2021. 1, 3, 6

[30] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik R Narasimhan, and Yuan Cao. React: Synergizing reasoning and acting in language models. In *The eleventh international conference on learning representations*, 2022. 2